

Spatial rescue in a biofilm under antibiotic treatment: a comprehensive project writeup

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1 What we are doing and why

1.1 The biological question

Bacteria living in biofilms are much harder to kill with antibiotics than free-floating (planktonic) cells. One proposed reason is purely physical: cells in the interior of a biofilm are shielded from drug diffusion, so only the surface layer experiences lethal concentrations. This creates a spatial refuge. But does that refuge actually matter for the population’s survival?

We study this through the lens of *evolutionary rescue*: the process by which a declining population avoids extinction because a resistant mutant arises and takes over before the population goes extinct. We ask whether the spatial structure of a biofilm makes rescue more likely than it would be in a well-mixed population of the same size.

The answer is not obvious. On one hand, the core is protected and could serve as a reservoir from which resistance can eventually emerge. On the other hand, only cells in the exposed edge can actually grow and mutate at the antibiotic-relevant rate—so having cells locked up in the core might simply reduce the number of cells available to generate the winning mutant. The outcome depends on whether the core is active (dividing) or dormant, and on the dose.

1.2 The model

We represent the biofilm as a one-dimensional chain of $N(t)$ cells ordered from the deep interior (index 0) to the outer surface (index $N - 1$). The chain has two compartments:

- **Edge:** the outermost l cells. These are exposed to antibiotic. Wild-type (WT) cells here have birth rate $b_e = 1$ (which sets the time unit) and death rate $d_e > 1$. We call d_e the *dose* and write $s_e \equiv d_e - b_e = \text{dose} - 1 > 0$ for the net excess death rate: this is the effective antibiotic selection pressure against WT.
- **Core:** all cells deeper than l from the surface, i.e. cells 0 through $N - l - 1$. Core cells are shielded from antibiotic. We set $b_c = d_c$ (neutral, no net growth). The default is $b_c = d_c = 0.2$, representing a metabolically active but slow-growing interior.

Events are drawn by the Gillespie algorithm. The critical coupling between compartments is mechanical: every edge birth pushes the innermost edge cell inward (into the core), and every edge death draws the outermost core cell outward (into the edge). This *conveyor-belt flow* means the core and edge are not isolated—cells continuously cycle between them as the edge population fluctuates.

Two phases of the trajectory. Under antibiotic treatment the edge is net-dying ($s_e > 0$), so the core-edge boundary retreats inward at average speed $s_e \cdot l$ per unit time. This gives the population dynamics two qualitatively distinct phases:

- **Phase 1** ($N > l$, core present): the boundary retreats at constant speed $s_e l$, so the total population declines *linearly*: $N(t) \approx N_0 - s_e l t$. This is because only the l edge cells drive the net decline; as long as the core exists, it feeds new cells into the edge at the same rate they are dying there. Phase 1 ends after $T_1 = (N_0 - l)/(s_e l)$ time units, when the last core cell is exposed.
- **Phase 2** ($N \leq l$, core exhausted): the core is gone and the entire remaining population is in the edge. Without resupply, the population declines *exponentially* at rate s_e per cell, exactly as a well-mixed population would.

The transition from Phase 1 to Phase 2 is the moment when the biofilm’s spatial structure disappears.

Resistance and rescue. Resistance (R) arises by mutation: with probability μ per replication event, a daughter is assigned a new, fully resistant genotype. Resistant cells divide at rate $b_R = 1$ (same as WT) and die at rate $d_R = 0.6$ in the edge, giving a net growth rate $s_R = b_R - d_R = 0.4 > 0$. In the core, R cells are subject to the same neutral dynamics as core WT ($b_c = d_c$). Rescue is declared when the number of R cells in the edge reaches the threshold $r^* = 10$.

1.3 Comparison design

We compare three conditions at every parameter point:

1. **Biofilm** ($l = 100$): the spatial model as described above.
2. **chainWM** ($l = 2000 \gg N_0 = 1000$): the chain model with an edge wider than the whole population, so the core never forms. This is the chain model’s own well-mixed limit.
3. **wellMixed**: an independent explicit well-mixed birth-death model using only the edge rates.

Conditions 2 and 3 must give statistically identical rescue probabilities (this is an implementation correctness check). The scientific comparison is biofilm vs. chainWM/wellMixed.

2 Analytical results

We now work through the three factors that determine rescue probability: how many mutations the biofilm produces (*mutation supply*), whether core-born mutations survive long enough to reach the edge (*delivery*), and whether any mutation can grow from a single cell to $r^* = 10$ once it gets there (*establishment*). Each factor on its own does not tell the full story; combining all three does.

Unless otherwise noted, all numerical results use the default parameters: $N_0 = 1000$, $l = 100$, $b_c = d_c = 0.2$, $b_e = b_R = 1$, $d_R = 0.6$ ($s_R = 0.4$), $\mu = 10^{-4}$, $r^* = 10$. Dose is swept from 1.05 to 2.85, corresponding to s_e from 0.05 to 1.85. All analytical results below are derived under the neutral-core assumption $b_c = d_c$.

2.1 Mutation supply

The most direct potential advantage of a biofilm is simple: the shielded core prolongs the population's lifetime, contributing extra cell divisions to the mutation pool.

Well-mixed baseline. A well-mixed population of N_0 cells declining at rate s_e per cell produces a total of $b_e N_0 / s_e$ replication events before extinction (integrating the birth rate over the exponential trajectory). Multiplying by μ :

$$\Lambda_{\text{WM}} = \frac{\mu b_e N_0}{s_e}. \quad (1)$$

In the rare-mutation limit ($\Lambda \ll 1$), $P_{\text{WM}} \approx \Lambda_{\text{WM}}$; more generally, $P_{\text{WM}} = 1 - e^{-\Lambda_{\text{WM}}}$.

Biofilm. The biofilm adds births from the core during Phase 1. Integrating the per-birth mutation probability over both phases:

$$\Lambda_{\text{BF}} = \frac{\mu}{s_e} \left[\frac{b_c (N_0 - l)^2}{2l} + b_e N_0 \right]. \quad (2)$$

The first term is the core contribution during Phase 1 (the factor $(N_0 - l)^2 / 2l$ is the total core births integrated over the linearly declining core size); the second is the combined edge contribution from both phases. The ratio simplifies to:

$$\frac{\Lambda_{\text{BF}}}{\Lambda_{\text{WM}}} = 1 + \frac{b_c (N_0 - l)^2}{2 b_e N_0 l}. \quad (3)$$

For default parameters this is $1 + 0.2 \times 900^2 / (2 \times 1 \times 1000 \times 100) \approx 1.81$.

Figure 1 makes two things clear. First, the mutation-supply advantage is *linear in b_c and entirely dose-independent*: it is determined by how fast the core divides relative to the edge, not by how quickly the antibiotic is working. Second, and more surprisingly, a *dormant core* ($b_c = 0$) provides no advantage at all. Setting $b_c = 0$ in Equation (2):

$$\Lambda_{\text{BF}}|_{b_c=0} = \frac{\mu b_e (N_0 - l)}{s_e} + \frac{\mu b_e l}{s_e} = \frac{\mu b_e N_0}{s_e} = \Lambda_{\text{WM}}. \quad (4)$$

A dormant core does extend the population's lifetime through Phase 1 (core cells survive but cannot divide), but it contributes zero births and therefore zero mutations. The longer lifetime and the reduced per-time mutation rate cancel exactly. *Passive protection alone is not enough*: the core must be metabolically active to contribute to

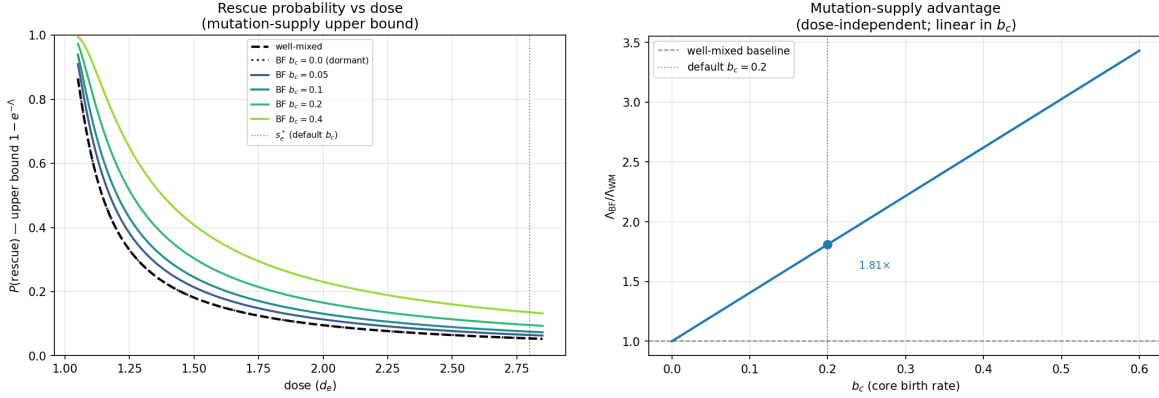


Figure 1: *Left:* Upper bound $1 - e^{-\Lambda}$ on rescue probability vs. dose, for several values of b_c . The dashed black line is the well-mixed baseline; the dotted $b_c = 0$ curve lies exactly on top of it. *Right:* Mutation-supply ratio $\Lambda_{\text{BF}}/\Lambda_{\text{WM}}$ vs. b_c . The ratio is linear in b_c and completely independent of dose.

rescue through mutation supply. At our default parameters the advantage is $1.81\times$ —real, but modest.

These curves are upper bounds because they treat every mutation as equally likely to rescue the population. The next two sections quantify the two additional hurdles—delivery and establishment—that make the true rescue probability far lower.

2.2 Core delivery: the race between drift and boundary retreat

A resistant mutation born deep in the core finds itself in a neutral environment ($b_c = d_c$) with no growth advantage. It must survive stochastic extinction long enough for the retreating boundary to expose it to the antibiotic zone.

Survival to delivery. A lineage born at depth d cells from the current boundary (i.e. at position $N_0 - l - d$) must wait roughly $\tau_d = d/(s_e l)$ time units before the boundary reaches it. During this wait, it undergoes a neutral birth-death process ($b_c = d_c$), which gives an extinction probability that grows with time: a single cell survives to time τ with probability $1/(1 + b_c \tau)$. This is the probability that the lineage is still alive when it is finally exposed to antibiotic.

Averaging over all birth depths (uniformly distributed from $d = 0$ to $d = N_0 - l - 1$):

$$\langle P(\text{survive to delivery}) \rangle = \frac{\ln(1 + \alpha)}{\alpha}, \quad \alpha \equiv \frac{b_c(N_0 - l)}{s_e l}. \quad (5)$$

The dimensionless parameter α is the ratio of core turnover speed to boundary retreat speed. Intuitively: if $\alpha \ll 1$ (fast boundary, slow drift), the boundary sweeps through

before drift can eliminate most lineages, and $\ln(1 + \alpha)/\alpha \approx 1$ —almost all lineages survive to delivery. If $\alpha \gg 1$ (slow boundary, fast drift), lineages wait so long that nearly all go extinct before delivery, and $\ln(1 + \alpha)/\alpha \approx \ln \alpha/\alpha \rightarrow 0$.

Net delivery rate. Multiplying the mutation supply rate by the survival probability:

$$\Gamma_{\text{core}} = \mu s_e l \ln(1 + \alpha). \quad (6)$$

This grows *logarithmically* in b_c (through α): increasing core turnover generates more mutations but also kills more lineages through faster drift, and the two effects nearly cancel.

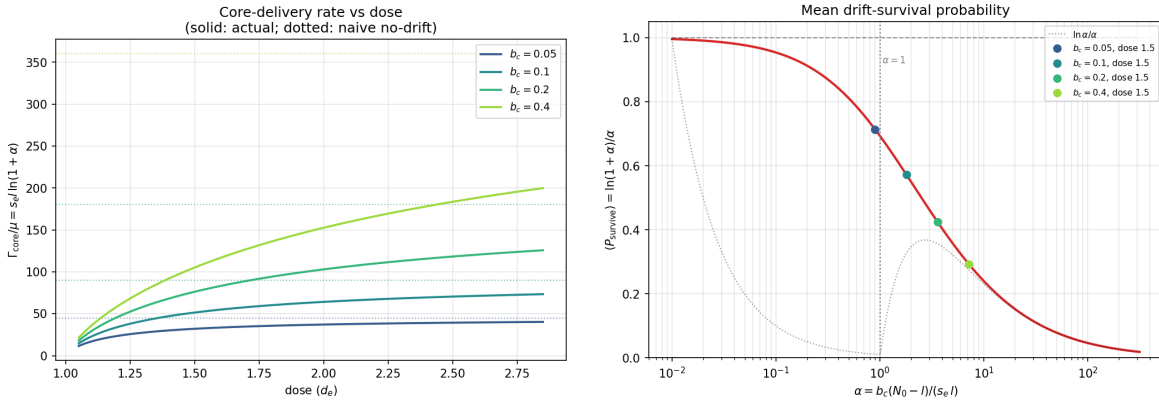


Figure 2: *Left:* Core-delivery rate Γ_{core}/μ vs. dose. Solid curves: actual rate accounting for drift losses (Equation (6)). Dotted horizontal lines: naive upper limit if every core mutation survived to delivery. The gap between them is the fraction lost to drift; it is largest at low dose (slow boundary retreat, large α). *Right:* Drift-survival probability $\ln(1 + \alpha)/\alpha$ vs. α . Below $\alpha = 1$ (left of the dotted vertical line) the core is an efficient nursery; above $\alpha = 1$ most lineages are lost before delivery.

Figure 2 makes the losses concrete. At low dose (large α , slow boundary retreat), survival collapses toward zero: lineages wait so long that drift eliminates nearly all of them. The crossover between the two regimes occurs at $\alpha = 1$:

$$s_e^* = \frac{b_c(N_0 - l)}{l}. \quad (7)$$

For default parameters, $s_e^* = 0.2 \times 900/100 = 1.8$ (dose 2.8), near the top of our swept range. Most of our simulated dose points fall in the $\alpha > 1$ regime where drift is actively suppressing core delivery.

There is an irony embedded in this result: the same antibiotic pressure that kills the population also drives the fast boundary retreat that makes the core useful. High selection is simultaneously the threat and the delivery mechanism.

2.3 Establishment: WT resupply keeps the edge hostile

Even a lineage that successfully reaches the edge must still grow from one (or a few) cells to the rescue threshold $r^* = 10$ before it is displaced or dies. This step is much harder in the biofilm than in the well-mixed model, because of what is happening to the WT population on the other side of the race.

Well-mixed. In the well-mixed model, as WT cells die, the competitive pressure on any R cell decreases over time. A single R cell growing at net rate $s_R = 0.4$ in a shrinking WT background has an establishment probability (probability of reaching $r^* = 10$) of approximately $p_{\text{est}}(1) = 1 - d_R/b_R = 0.40$.

Biofilm Phase 1: WT resupply. During Phase 1 something qualitatively different happens. Every time a WT cell in the edge dies, the outermost core cell—which is WT throughout most of Phase 1—immediately moves into the edge to fill the vacancy. The WT count in the edge is actively *maintained at l cells* by the conveyor belt. Antibiotic kills WT cells at the edge, but the core continuously replaces them. The competitive pressure on R never weakens.

To compute the establishment probability we write a Markov chain for k , the number of R cells in the edge ($0 \leq k \leq r^*$). Under a uniform-mixing approximation (each cell in the edge is equally likely to be chosen for displacement), the transition rates are:

$$q_{k \rightarrow k+1} = b_R \cdot k \cdot \frac{l-k}{l} \quad (\text{R birth displaces a WT cell inward}) \quad (8)$$

$$q_{k \rightarrow k-1} = k \left[b_e \cdot \frac{l-k}{l} + d_R \right] \quad (\text{WT birth displaces R inward, or R dies and WT enters from core}). \quad (9)$$

An R birth and a WT birth each displace a cell of the opposite type at the same rate (proportional to the fraction of the opposing type). Additionally, R deaths bring in a fresh WT cell from the core regardless of b_e . This is the Gambler's ruin problem with state-dependent rates; it can be solved exactly for small r^* .

Figure 3 shows the result: $p_{\text{est}}^{\text{BF}}(1) \approx 0.0048$ vs. $p_{\text{est}}^{\text{WM}}(1) = 0.40$ —an 83-fold reduction.

The mechanism is clean. Consider a single R cell in the biofilm edge. Its effective net growth rate, accounting for the displacement dynamics of both births and deaths, is $b_R - b_e - d_R = 1.0 - 1.0 - 0.6 = -0.6$: it is mildly *deleterious*. R cells die less often than WT (that is their resistance), but since $b_R = b_e = 1$, they cannot win edge slots through births any faster than WT can reclaim them via core resupply. The core continuously undoes the selective advantage that the antibiotic is supposed to provide.

Two additional properties are worth noting. First, this suppression is *dose-independent*: the transition rates in Equations (8) and (9) depend on b_R , b_e , d_R , and l , but not on s_e or dose. WT resupply operates at the same intensity regardless of how aggressively

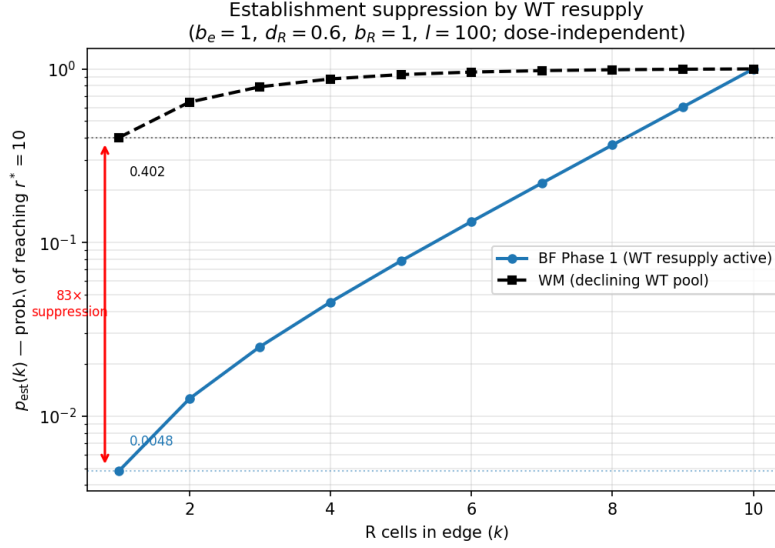


Figure 3: Establishment probability $p_{\text{est}}(k)$ —the probability of reaching $r^* = 10$ starting from k R cells in the edge—for the biofilm Phase 1 chain with WT resupply (blue) and the well-mixed declining population (black dashed). The suppression at $k = 1$ is 83-fold. This suppression is dose-independent.

the antibiotic is working. Second, it occurs even for a *dormant core* ($b_c = 0$): core cells move into the edge upon WT deaths regardless of whether they divide, so the resupply flux $d_e \cdot l$ is driven by WT deaths, not core births.

One important caveat: the 83-fold figure assumes uniform mixing within the edge. In the actual simulation, births occur adjacent to the parent, and R cells may cluster near the outer surface where displacement by WT is least likely. The true suppression is probably weaker, but by how much requires simulation to determine.

2.4 Putting it together: establishment suppression dominates

We now have quantitative estimates for all three factors. In the rare-mutation limit, the total rescue probability for the biofilm is approximately:

$$P_{\text{rescue}}^{\text{BF}} \approx \underbrace{\Lambda_{P1} \cdot p_{\text{est}}^{\text{BF}}}_{\text{Phase 1 edge}} + \underbrace{\Lambda_{P2} \cdot p_{\text{est}}^{\text{WM}}}_{\text{Phase 2 edge}} + \underbrace{P_{\text{core}}}_{\text{core delivery}}. \quad (10)$$

Here Λ_{P1} and Λ_{P2} are the mutation supply contributions from the edge during Phase 1 and Phase 2 respectively; mutations arising in Phase 2 face no WT resupply (the core is exhausted by then) and establish with the well-mixed probability $p_{\text{est}}^{\text{WM}}$. The core-delivery term P_{core} integrates over all birth depths, weighting by drift survival probability and the bolus-size distribution, and applies the Gambler's ruin result $p_{\text{est}}(k)$ for the delivered number of cells.

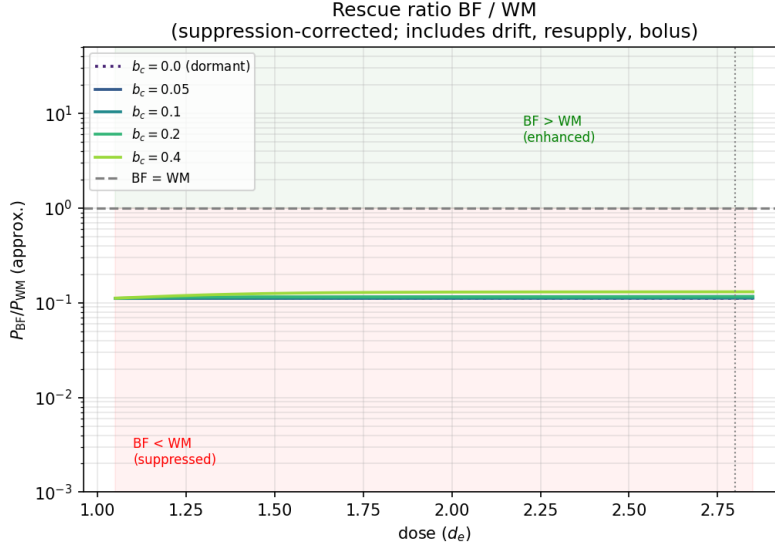


Figure 4: Approximate rescue probability ratio $P_{\text{BF}}/P_{\text{WM}}$ vs. dose, for five values of b_c . The ratio is ≈ 0.11 across the entire dose range—nearly flat and well below 1 for all parameters shown. Red shading marks the suppression regime (< 1); green marks enhancement (> 1), which is not reached here.

Figure 4 shows the result. Across the entire swept dose range and for all b_c values, $P_{\text{BF}}/P_{\text{WM}} \approx 0.11$: the biofilm rescues roughly one-ninth as often as the well-mixed control. The curves are nearly flat in dose and barely sensitive to b_c .

The dose-independence has a clean analytical explanation. A fraction $(N_0 - l)/N_0 = 0.9$ of all edge mutations arise during Phase 1 (the Phase 1 duration scales as $1/s_e$, but so does the total mutation supply, so the *fraction* is dose-independent). These mutations all face the 83-fold establishment suppression. The remaining 0.1 fraction arises in Phase 2 with the well-mixed establishment probability. Combining:

$$\frac{P_{\text{BF}}^{\text{edge}}}{P_{\text{WM}}} \approx 0.9 \times \frac{0.0048}{0.40} + 0.1 \approx 0.11. \quad (11)$$

Neither the 0.9/0.1 split nor $p_{\text{est}}^{\text{BF}}$ depends on dose. The core-delivery term adds a small dose-dependent contribution at high dose (faster boundary retreat delivers larger boluses) but is insufficient to push the ratio above 1 within our swept range.

The core conclusion: the $1.81\times$ mutation-supply advantage from Section ?? is overwhelmed by the $83\times$ establishment suppression. The upper-bound curves in Figure 1 are systematically misleading—extra mutations that cannot establish do not contribute to rescue.

2.5 Testable predictions

The analysis yields several falsifiable predictions for the simulation:

1. The biofilm should rescue *less* often than the well-mixed control across the swept dose range, with ratio ≈ 0.11 . This ratio is a lower bound; spatial clustering of R cells near the outer edge could increase $p_{\text{est}}^{\text{BF}}$ and raise the true ratio.
2. The ratio $P_{\text{BF}}/P_{\text{WM}}$ should be nearly flat in dose. Varying dose should not substantially change the relative rescue probability.
3. Varying b_c should fan out the rescue curves at high dose (where core delivery becomes efficient, $\alpha < 1$) while all curves converge near the well-mixed baseline at low dose ($\alpha \gg 1$, delivery suppressed by drift).
4. The crossover dose s_e^* scales linearly with b_c (Equation (7)), so increasing b_c shifts the dose at which the biofilm starts to show a core-delivery advantage.
5. In the rare-mutation regime, $P(\text{rescue}) \propto \mu$: changing μ shifts the curves vertically on a log scale but does not change their shape.

Whether prediction 1 holds in practice—and specifically, whether spatial clustering is strong enough to overcome the 83-fold analytical suppression and put $P_{\text{BF}} > P_{\text{WM}}$ —is the central empirical question the simulation is designed to answer.

2.6 What the approximation misses

- **Uniform mixing within the edge.** R cells are assumed to be uniformly distributed among the l edge positions. In the actual simulation, births occur adjacent to the parent, producing spatial clustering near the outer surface where displacement risk is lowest. This approximation is expected to overestimate the 83-fold suppression.
- **Independent lineages.** Two R lineages can compete if both are in the edge simultaneously. At $\mu = 10^{-4}$ and $N_0 = 1000$, $\Lambda \lesssim 1$ across most of our range, so clonal interference is small but not negligible.
- **Bolus delivery.** Surviving core lineages arrive as multi-cell clumps of expected size $1 + b_c\tau$ (a lineage that survived to time τ has had $\approx b_c\tau$ births by then). Since $p_{\text{est}}(k)$ is convex in k , larger boluses help superlinearly—a 3-cell bolus does better than three independent 1-cell arrivals. This is partially accounted for in P_{core} but the bolus-size distribution is approximate.
- **Uniform birth-depth distribution.** Core-born mutations are assumed equally likely to arise at any depth. In reality births are concentrated near the boundary where cell turnover is highest.

3 What we have built

3.1 Simulation code

`src/chainModel.py`: Full Gillespie simulation of the 1D chain model. Tracks every resistant lineage independently with metadata: birth compartment, time, whether it ever reached the edge. Records a trajectory snapshot every 50 events (for visualization). At rescue, takes a snapshot of edge lineage counts before continuing (needed because post-rescue dynamics would corrupt measurements). Returns: rescue/extinction outcome, timing, rescue mode ('core' or 'edge' based on origin of the dominant edge lineage at rescue), lineage appearance/extinction/survival counts per compartment.

`src/wellMixed.py`: Independent well-mixed birth-death model using only edge rates. No spatial structure. Used as both a scientific baseline and an implementation cross-check (chainModel with $l \geq N_0$ must match it statistically).

`src/kymograph.py`: Visualization of a single simulation run as a space-time heatmap. WT cells gray; resistant lineages colored by tab20 colormap (rescuing lineages saturated, non-rescuers desaturated). Renders the conveyor-belt dynamics directly.

3.2 Run scripts

`scripts/runJobs/runBatch.py`: Runs one batch of simulations for a single parameter point (one condition, one dose, one seed range). Writes a CSV row per run. Designed to be called from HTCondor job arrays on OSG.

`scripts/runJobs/quickSweep.py`: Runs a local validation sweep (3 conditions $\times$ 8 doses $\times$ 200 seeds $\approx$ 6 min). Produces a summary plot with rescue curves, pathway breakdown, and chainWM/wellMixed agreement check.

`scripts/aggregateResults.py`: Aggregates CSVs from a full OSG sweep and produces publication-quality figures.

3.3 Make-jobs and HTCondor scripts

`scripts/makeJobs/genSweepMainJobs.py`: Generates argument lists for the main sweep (300 jobs: 3 conditions $\times$ 10 doses $\times$ 10 seed batches).

`scripts/makeJobs/genCoreSensitivityJobs.py`: Generates argument lists for the b_c sensitivity sweep (135 jobs: 3 b_c values $\times$ 3 conditions $\times$ 15 doses $\times$ 3 batches).

`scripts/makeJobs/genMuSensitivityJobs.py`: Generates argument lists for the μ sensitivity sweep (540 jobs: 4 μ values $\times$ 3 conditions $\times$...).

Each sweep has a corresponding `.sub` HTCondor submission file in `scripts/runJobs/`. Results land in `results/`, logs in `logs/`.

3.4 Tests

`src/testTracking.py`: Tests internal consistency of the chain model. Verifies lineage accounting (appearances, extinctions, survivors sum correctly), compartment-size invariants, and that the return dict has all expected keys. Runs fast (~ 5 seeds); assertions fire on violation.

`scripts/testEquiv.py`: Statistical equivalence test. Runs 300 seeds for `chainModel(l = 2000)` and `wellMixed` at several doses and prints side-by-side rescue probabilities. Visible divergence signals a bug.

4 What needs to get done

4.1 Simulations to run

1. **Quick local sweep** (already working): run `quickSweep.py` to confirm the pipeline is clean and `chainWM/wellMixed` agree before submitting to OSG.
2. **Main OSG sweep**: 3 conditions $\times$ 10 doses $\times$ ~ 1000 seeds per point. Expected runtime: hours on OSG.
 - Key observable: is $P_{\text{BF}} < P_{\text{WM}}$ as the analytical picture predicts? Does the ratio flatten at ≈ 0.11 , or does spatial clustering push it higher?
 - Does the crossover dose match $s_e^* = 1.8$?
3. **Core-rate sensitivity sweep** (b_c/b_e varied): Fan-out test. Rescue curves for different b_c should converge near $s_e \rightarrow 0$ and diverge at $s_e > s_e^*$. The dose of separation should shift with b_c per Equation (7).
4. **Mutation-rate sweep** (μ varied): Verify $P(\text{rescue}) \propto \mu$ in the rare-mutation regime. Four values spanning 10^{-6} to 10^{-3} .
5. **Dormant-core limit test**: Run biofilm condition with $b_c = d_c = 0$ and compare to `wellMixed` directly. This is the cleanest test of Equation (4). *Not yet done.*

4.2 Analysis to build

1. **Overlay Λ curves on rescue probability plots**: Plot $1 - e^{-\Lambda_{\text{BF}}}$ and $1 - e^{-\Lambda_{\text{WM}}}$ directly on top of the simulation rescue curves. This visualizes the gap between the analytical upper bound and the true probability, showing where the suppression effects (drift, resupply) live.
2. **Pathway fraction plot**: fraction of biofilm rescues attributed to core-delivery vs. edge-mutation, as a function of dose. Expected: core-delivery fraction rises with dose (edge mutants become less viable as WT resupply strength is constant

but population declines faster). Tracked by `rescueMode` in the simulation output and already plumbed into `aggregateResults.py`.

3. **Γ_{core} overlay:** overlay the predicted core-delivery rate (Equation (6)) on the observed core-pathway fraction. Direct test of the logarithmic saturation prediction.
4. **Lineage survival curves:** from the simulation output, plot empirical delivery survival probabilities as a function of birth depth (uses the lineage metadata already tracked). Compare to Equation (5).

4.3 Theory still needed

1. **Empirical $p_{\text{est}}^{\text{BF}}$:** measure it directly from simulation as the fraction of single-cell edge appearances that grow to r^* . This will reveal how much the uniform-mixing approximation overestimates the suppression, and whether the true $p_{\text{est}}^{\text{BF}}$ is consistent with $P_{\text{BF}}/P_{\text{WM}} > 1$ being possible.
2. **Full rescue probability formula:** combine Λ_{BF} , Γ_{core} , and the empirically corrected $p_{\text{est}}^{\text{BF}}$ into a single closed-form approximation that can be plotted alongside simulation curves.
3. **Bolus delivery quantification:** estimate the nonlinear advantage of multi-cell core arrivals from simulation (fraction of core rescues with $k > 1$ at delivery). Determine whether modeling the bolus size distribution explicitly improves the prediction.
4. **Finite- μ corrections:** at $\mu = 10^{-3}$ with $N_0 = 1000$, $\Lambda \approx 10$, and multiple lineages arise simultaneously. The rare-mutation approximation breaks down and clonal interference must be accounted for.

4.4 Writing

1. Fill in `derivation.tex` Sections 4 (Simulation results) and 5 (Discussion) once sweeps are complete.
2. Compare to Fruet et al. 2025 and other biofilm-rescue literature.
3. Decide whether `derivation.tex` is the target manuscript or whether results sections should move to a separate paper.

5 Parameter reference

Parameter	Symbol	Default	Description
Initial population	N_0	1000	Total cells at $t = 0$
Edge width	l	100	Antibiotic-exposed cells
Core width		900	$N_0 - l$
WT edge birth	b_e	1.0	Sets time unit
WT edge death	d_e	dose	Swept variable (> 1)
WT edge selection	s_e	dose-1	Net WT death rate in edge
WT core birth/death	$b_c = d_c$	0.2	Neutral core
R edge birth	b_R	1.0	Same as WT
R edge death	d_R	0.6	Sublethal; $s_R = b_R - d_R = 0.4 > 0$
Mutation rate	μ	10^{-4}	Per birth event
Rescue threshold	r^*	10	R edge cells required for rescue
Biofilm edge width	l	100	Main simulation condition
chainWM edge width	l	2000	$\gg N_0$; core never forms
Dose sweep	s_e	0.05–1.85	10 points, log-spaced
b_c sweep	b_c/b_e	{0.05, 0.1, 0.2}	Core-rate sensitivity
μ sweep	μ	{ 10^{-6} , 10^{-5} , 10^{-4} , 10^{-3} }	

Crossover parameter (default): $\alpha \equiv b_c(N_0 - l)/(s_e l)$; $\alpha = 1$ at $s_e^* = 0.2 \times 900/100 = 1.8$ (dose = 2.8).

Mutation-supply ratio (default): $\Lambda_{\text{BF}}/\Lambda_{\text{WM}} = 1 + b_c(N_0 - l)^2/(2b_e N_0 l) \approx 1.81$.